Mini Project: Evaluating Clustering Before and After PCA

Objective

Understand the impact of dimensionality reduction on clustering performance by comparing results before and after applying PCA.

Task Description

You are required to:

1. Use a dataset with multiple features (**Wine Dataset is recommended**).
2. Apply clustering before and after dimensionality reduction.
3. Compare and analyze the results.

Suggested Dataset

You may use:

- **Wine Dataset (recommended)**

  (available in sklearn: `sklearn.datasets.load_wine` *)*

💡 You may also use your own dataset (must contain multiple numerical features).

Steps

1. **Preprocess data**
   - Handle missing values (if any)
   - Apply scaling (Standardization is required)
2. **Apply Clustering (Before PCA)**
   - Apply **K-Means** on the original data
   - Record results

3. **Apply PCA**
    - Reduce the data to **2 components**
4. **Apply Clustering (After PCA)**
    - Apply **K-Means** on the transformed data
5. **Visualization**
    - Visualize clusters (2D using PCA components)
    - Compare before vs after PCA

Analysis (Very Important!)

You must answer:

- How did PCA affect clustering results?
- Did visualization become clearer?
- Did you lose important information?
- Which approach is better and why?

Deliverables

- Jupyter Notebook (.ipynb) including:
    - Code
    - Visualizations
    - Clear explanations
- Final summary (Key Insights)

Bonus (Optional)

- Try different number of PCA components
- Compare results using another clustering algorithm (e.g., DBSCAN)